

- 6 Mary and Jane are standing at each end of an ice board 1.0 m in length. The ice board rests on frictionless ice. Mary throws a 5.0 kg ball horizontally to Jane who catches it.

Given that the mass of Mary, Jane and the ice board are 42 kg, 40 kg and 3.0 kg respectively, what is the distance travelled by the ice board before coming to rest again?

- A 0.053 m B 0.056 m C 0.059 m D 0.063 m

By principle of conservation of linear momentum,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$0 = 5.0u - (42 + 40 + 3.0)v$$

$$u = 17v$$

Time of flight, t ,

$$t = \frac{L}{u + v}$$

Distance travelled by ice board, d ,

$$d = vt$$

$$= \frac{Lv}{u + v}$$

$$= \frac{Lv}{17v + v}$$

$$= \frac{L}{17 + 1}$$

$$= \frac{1.0}{18}$$

$$= 0.056 \text{ m}$$

Ans: B

- 7 A corker is initially at rest on the cocked gun shown in A in the diagram below. It then rises